VOTING ELIGIBILITY SYSTEM

Python 3 Mini Project Report

Language: Python 3

1. Problem Statement

Develop a console-based program that accepts a person's name, age, and Indian citizenship status, then determines whether the person is eligible to vote. The program must validate age input, explain ineligibility, support checking multiple voters, and display final totals.

2. Objectives

• Check voting eligibility using age and citizenship.
• Validate age as a whole number from 0 to 120.
• Use try-except to handle invalid numeric input.
• Use functions, conditions, and a loop.
• Track and display eligible and ineligible voter totals.

3. Eligibility Rules

• Eligible: age is 18 or more and citizenship is y.
• Under 18: not eligible; show years remaining (18 – age).
• Age 18 or more with citizenship n: not eligible because the person is not an Indian citizen.
• If both conditions fail, display both reasons.

4. Concepts Used

Functions: Separate input validation, citizenship validation, eligibility checking, and program flow.
Conditions: if-elif-else checks age and citizenship.
Exceptions: try-except catches non-integer age input.
Loops: while loops re-prompt invalid entries and allow multiple voters.
Counters: Track total, eligible, and ineligible voters.

5. Algorithm

1. Start the program and initialize all counters to zero.
2. Ask for the voter's name.
3. Ask for age until it is an integer between 0 and 120.
4. Ask for citizenship until the answer is y or n.
5. Check eligibility and print the result and reason if needed.
6. Update the counters.
7. Ask whether to check another voter; repeat if the answer is y.
8. Print the final summary and stop.

6. Source Code

```
def get_valid_age():
    """Ask repeatedly until the user enters an integer from 0 to 120."""
    while True:
        age_text = input("Age: ").strip()
        try:
            age = int(age_text)
        except ValueError:
            print("Invalid input. Enter a whole number.")
            continue

        if age < 0 or age > 120:
            print("Invalid age. Age must be between 0 and 120.")
            continue

        return age

def get_citizenship():
```

```
    """Accept only y or n for Indian citizenship."""
    while True:
        citizen = input("Indian citizen (y/n): ").strip().lower()
        if citizen in ("y", "n"):
            return citizen
        print("Invalid input. Please enter y or n.")


def check_eligibility(name, age, citizen):
    """Print the person's voting eligibility and return True if eligible."""
    if age >= 18 and citizen == "y":
        print(f"{name} is eligible to vote.")
        return True
    elif age < 18 and citizen == "y":
        print(f"{name} is not eligible to vote.")
        print(f"Reason: Under 18. Years left to become eligible: {18 - age}")
        return False
    elif age >= 18 and citizen == "n":
        print(f"{name} is not eligible to vote.")
        print("Reason: Not an Indian citizen.")
        return False
    else:
        print(f"{name} is not eligible to vote.")
        print(f"Reason: Under 18 and not an Indian citizen. Years left to become eligible: {18 - age}")
        return False


def main():
    voters_checked = 0
    eligible_count = 0
    not_eligible_count = 0

    print("=== Voting Eligibility System ===")

    while True:
        name = input("\nName: ").strip()
        while not name:
            print("Name cannot be empty.")
            name = input("Name: ").strip()

        age = get_valid_age()
        citizen = get_citizenship()

        voters_checked += 1
        if check_eligibility(name, age, citizen):
            eligible_count += 1
        else:
            not_eligible_count += 1

        another = input("\nCheck another voter? (y/n): ").strip().lower()
        while another not in ("y", "n"):
            print("Invalid input. Please enter y or n.")
            another = input("Check another voter? (y/n): ").strip().lower()

        if another == "n":
            break

    print("\n=== Final Summary ===")
    print(f"Voters checked: {voters_checked} | Eligible: {eligible_count} | Not eligible: {not_eligible_count}")


if __name__ == "__main__":
    main()
```

7. Test Cases

1. Normal input

Input: Aman; age 25; citizen y
Expected output: Aman is eligible to vote. Summary after one voter: checked 1, eligible 1, not eligible 0.

2(a). Boundary age

Input: Neha; age 18; citizen y
Expected output: Neha is eligible to vote.

2(b). Underage

Input: Ravi; age 17; citizen y
Expected output: Ravi is not eligible to vote. Years left to become eligible: 1.

3. Invalid age

Input: Amit; age abc, then -5, then 30; citizen y
Expected output: Shows invalid whole-number message for abc, range message for -5, then accepts 30 and reports Amit is eligible to vote.

4. Non-citizen

Input: Riya; age 30; citizen n
Expected output: Riya is not eligible to vote. Reason: Not an Indian citizen. Summary after one voter: checked 1, eligible 0, not eligible 1.

9. Conclusion

The project demonstrates the use of Python functions, conditional statements, loops, exception handling, input validation, and counters to build a simple voting eligibility checker.